

AI-Native PPTX Generation: The DeckForge Vision

No templates. No placeholders. AI writes the deck. You send it.

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1. Executive Summary

DeckForge's ultimate goal is to eliminate manual presentation creation for recurring business deliverables like **Financial Due Diligence (FDD)**, **Tax Due Diligence (TDD)**, and **Internal Audit (IA)** proposals.

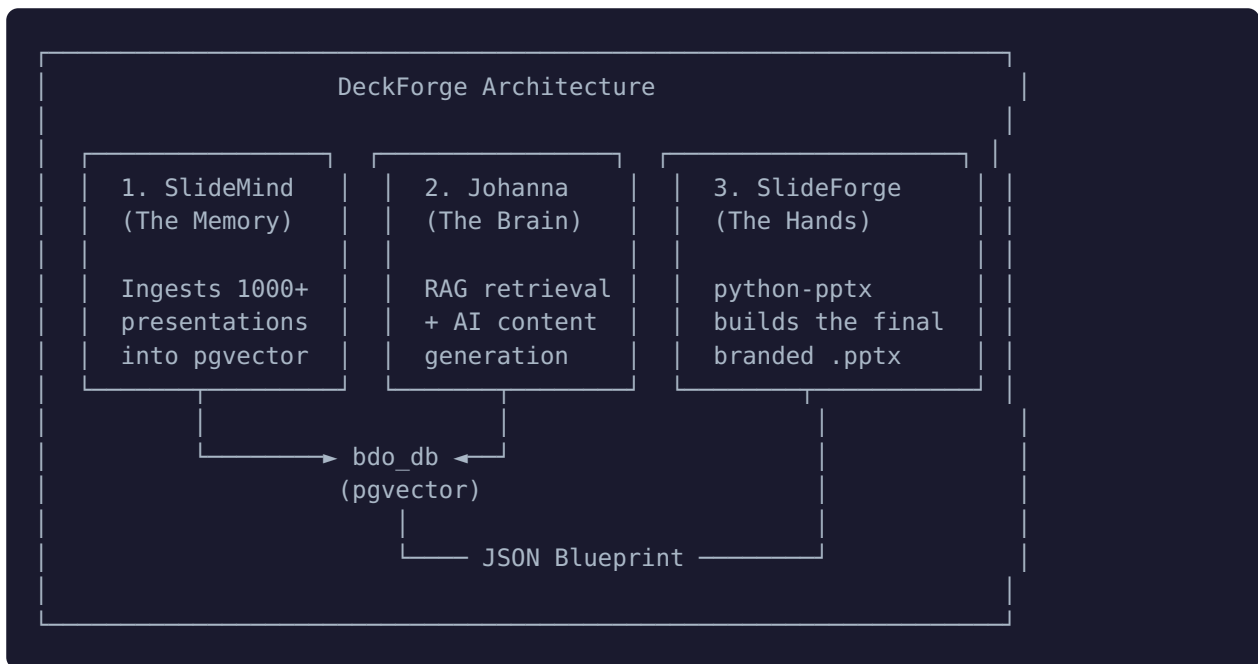
Instead of templates with `{{PLACEHOLDER}}` tags (old school), the AI learns from **1000+ historical presentations** and generates complete, branded, ready-to-send PPTX files from a **30-second human briefing**.

The Promise

Traditional (Old School)	AI-Native (DeckForge)
Copy an old deck, change client name	AI writes fresh, contextual content
Find-and-replace placeholders	AI decides structure, layout, text
2–4 hours of manual editing	30 seconds of input → ready deck
Inconsistent quality across proposals	AI applies distilled best practices
Knowledge trapped in individual files	Knowledge lives in a searchable RAG database

2. Architecture: The Three Engines

DeckForge is built on three autonomous engines that work together:



Engine 1: SlideMind — The Memory

Purpose: Extract, analyze, and vectorize every slide from every historic presentation.

What it does: - Scans thematic folders (/BD0/FDD/source/ , /BD0/TDD/source/) - Extracts text, styles, comments, and structure from each slide - Generates AI summaries per slide and per presentation - Stores everything in PostgreSQL with `pgvector` embeddings - Groups knowledge by tenant and theme (`SummarizeTheme`) - Produces thematic "seed" knowledge (merged best practices)

Current status: ✅ Operational — 22+ PPTX files processed and embedded in `bdo_db` .

Engine 2: Johanna — The Brain

Purpose: Understand the briefing, retrieve relevant knowledge, and generate a structured slide blueprint.

What it does: - Receives a natural language briefing (who, what, when) - Searches `bdo_db` via cosine similarity to find the most relevant past slides - Generates a **JSON Slide Blueprint** — a structured description of every slide that should appear in the new deck - Each slide in the blueprint contains: type, title, content, bullets, tables, and layout hints

Current status: ✅ RAG retrieval working (0.72–0.75 similarity scores for FDD queries).

Engine 3: SlideForge — The Hands

Purpose: Take the JSON blueprint and produce a real `.pptx` file using BDO's brand identity.

What it does: - Reads the JSON Slide Blueprint from Johanna - Opens a **BDO Master Layout** file (brand colors, fonts, logo — no content) - Creates each slide programmatically using `python-pptx` - Picks the right layout for each slide type (cover, bullet, table, team, timeline) - Exports the final `.pptx` ready to email

Current status: ☐ To be built.

3. The Workflow: From Briefing to Deck

Step 1: Human Briefing (30 seconds)

The consultant provides minimal input:

```
client: "Acme Corporation"
target: "MBH Bank - ATM Division"
service: "FDD + TDD"
industry: "Financial Services / Banking"
timeline: "6 weeks"
team_lead: "Dr. Kovács Péter"
language: "Hungarian"
special_notes: "Focus on regulatory compliance (MNB)"
```

Step 2: AI Knowledge Retrieval (Automatic)

Johanna queries `bdo_db` and retrieves: - Past FDD proposals for similar industries - Pricing patterns for 6-week engagements - Standard BDO methodology sections - Team structure examples for financial services - Regulatory compliance language from past MBH-related work

Step 3: AI Content Generation (Automatic)

The AI produces a **JSON Slide Blueprint**:

```
{
  "metadata": {
    "title": "Acme Corporation – Pénzügyi és Adóátvilágítás",
    "total_slides": 12,
    "language": "hu",
    "generated_at": "2026-02-21T15:00:00Z"
  },
  "slides": [
    {
      "number": 1,
      "type": "cover",
      "layout": "title_slide",
      "content": {
        "title": "Acme Corporation",
```

```

    "subtitle": "Pénzügyi és Adóátvilágítás (FDD+TDD)",
    "date": "2026. február",
    "confidential": true
  }
},
{
  "number": 2,
  "type": "agenda",
  "layout": "bullet_slide",
  "content": {
    "title": "Tartalomjegyzék",
    "bullets": [
      "A BDO bemutatása",
      "Az átvilágítás célja és hatóköre",
      "Módszertan és megközelítés",
      "Csapatösszetétel",
      "Ütemterv",
      "Díjazás"
    ]
  }
},
{
  "number": 5,
  "type": "methodology",
  "layout": "two_column",
  "content": {
    "title": "Módszertan",
    "left_column": {
      "heading": "Pénzügyi Átvilágítás (FDD)",
      "bullets": [
        "Pénzügyi kimutatások elemzése (3 év)",
        "Normalizált EBITDA meghatározása",
        "Net debt / working capital korrekciók",
        "Red Flag jelentés az első 2 hétben"
      ]
    },
    "right_column": {
      "heading": "Adóátvilágítás (TDD)",
      "bullets": [
        "ÁFA és társasági adó megfelelés",
        "Transzferár dokumentáció vizsgálat",
        "Adókockázatok számszerűsítése",
        "MNB szabályozási megfelelés"
      ]
    }
  }
},
{
  "number": 8,
  "type": "pricing",
  "layout": "table_slide",
  "content": {
    "title": "Díjazás",
    "table": {
      "headers": ["Szolgáltatás", "Óradíj (EUR)", "Becsült óraszám", "Összesen (EUR)"],
      "rows": [
        ["FDD – Partner", "350", "40", "14,000"],

```

```
[
  [
    ["FDD – Manager", "250", "120", "30,000"],
    ["TDD – Tax Partner", "350", "30", "10,500"],
    ["TDD – Senior", "200", "80", "16,000"]
  ],
  {
    "total": "70,500 EUR + ÁFA"
  }
]
```

Step 4: PPTX Assembly (Automatic)

SlideForge reads the blueprint, opens the BDO Master Layout, and builds each slide:

```
Input:  slide_blueprint.json + BDO_Master.pptx (brand only)
Output: Acme_Corporation_FDD_TDD_Ajanlat_2026.pptx (12 slides, ready to send)
```

4. Technology Stack

Layer	Technology	Purpose
Knowledge Store	PostgreSQL + pgvector	Vector similarity search over 1000+ decks
Embeddings	Gemini embedding-001	3072-dimension vectors for semantic search
AI Brain	Gemini 2.5 Flash/Pro	Content generation from RAG context
PPTX Builder	python-pptx	Programmatic slide creation with full control
Brand Source	BDO Master .pptx	Slide layouts, colors, fonts, logo
Orchestrator	Go (DeckForge CLI)	Ties everything together in one command
Chat Interface	Johanna	Interactive refinement and Q&A

Why python-pptx ?

- Full programmatic control over every shape, text run, font, color, and position
- Can create tables, charts, images, and complex layouts
- Works directly with `.pptx` XML structure — no LibreOffice needed
- Mature library, battle-tested for enterprise PPTX generation
- Can read Master Layout slide definitions and apply them automatically

5. The BDO Master Layout

The Master Layout is **not a template** — it contains **zero content**. It defines only the visual DNA:

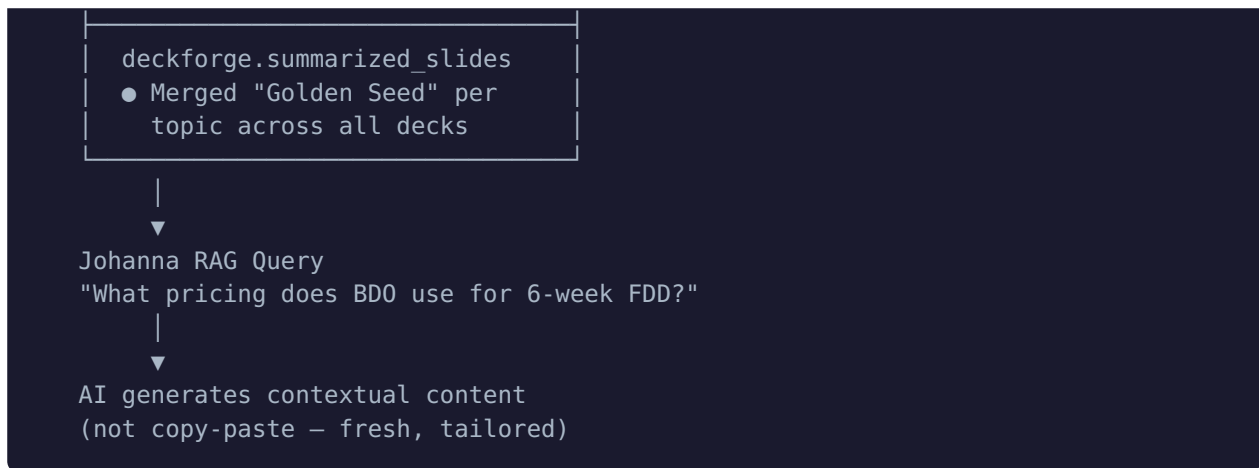
Layout Name	Purpose	Elements
title_slide	Cover page	BDO logo, title, subtitle, date
section_header	Chapter dividers	Large text, accent bar
bullet_slide	Standard content with bullets	Title + body placeholder
two_column	Side-by-side comparison	Title + 2 body areas
table_slide	Pricing, timelines, data	Title + table placeholder
team_slide	Team introduction	Photo placeholders + titles
closing_slide	Contact information	Partner photo, email, phone

Key: BDO's brand team provides this once. It never changes unless the brand identity changes.

6. Knowledge Architecture

How Knowledge Flows





Three Levels of Knowledge

1. **Raw Slides** — Individual slide text from each presentation (1000+ files × 10-15 slides each)
2. **Embeddings** — Vectorized representations for similarity search (what we built today)
3. **Golden Seeds** — AI-merged best practices per topic, distilled from hundreds of similar slides (via `slidemind summarize`)

7. Implementation Roadmap

Phase 1: Foundation (Done)

- [x] PPTX text extraction pipeline (`pptx_to_mcp.sh`)
- [x] Gemini embedding pipeline (`embed_pptx.sh`)
- [x] `bdo_db` with pgvector and tenant partitioning
- [x] RAG retrieval verified (0.72–0.75 similarity)
- [x] 22+ BDO presentations processed

Phase 2: Deep Knowledge (Next)

- [] Run `slidemind scan` on all 1000+ PPTX files (full ingestion with AI analysis)
- [] Run `slidemind summarize --tenant BDO --theme FDD` (create Golden Seeds)
- [] Run `slidemind summarize --tenant BDO --theme TDD`
- [] Verify knowledge granularity (slide-level vs. presentation-level)

Phase 3: AI Blueprint Generator

- [] Create FDD Generator MCP for Johanna (defines JSON Slide Blueprint schema)
- [] Train the prompt with examples from Golden Seeds

- [] Test blueprint generation: briefing → JSON → validate structure
- [] Add multi-language support (HU/EN)

Phase 4: PPTX Builder Engine

- [] Obtain/create BDO Master Layout `.pptx`
- [] Build `python-pptx` builder script (`scripts/forge_pptx.py`)
- [] Map JSON slide types → Master Layout slide layouts
- [] Handle tables, bullets, two-column, images
- [] Add cover slide with dynamic date and confidentiality notice

Phase 5: One-Command Pipeline

- [] Create `deckforge forge` CLI command
- [] Input: YAML/JSON briefing file or interactive prompts
- [] Output: Finished `.pptx` in `output/` directory
- [] Optional: Johanna chat-based interactive refinement
- [] Optional: PDF export alongside PPTX

Phase 6: Scale

- [] Process remaining 1000+ PPTX files
- [] Add TDD, IA, and Valuation service types
- [] Multi-tenant support (BDO Hungary, BDO Global)
- [] Quality scoring: AI rates its own output against past top-rated proposals
- [] Feedback loop: human corrections improve future generations

8. CLI Interface

Quick Generation

```
# Generate from a briefing file
deckforge forge --tenant BDO --theme FDD --brief briefing.yaml

# Interactive mode (AI asks the questions)
deckforge forge --tenant BDO --theme FDD --interactive

# Generate blueprint only (for review before building PPTX)
deckforge forge --tenant BDO --theme FDD --brief briefing.yaml --blueprint-only
```


Johanna Chat Mode

User: "Készíts egy FDD ajánlatot az Acme Corp részére, logisztikai iparág, 6 hetes projekt"
Johanna: [Retrieves RAG context] → [Generates JSON Blueprint] → [Calls SlideForge] →
"Az ajánlat elkészült: Acme_Corp_FDD_Ajanlat_2026.pptx (12 slide, magyar)"

9. Success Criteria

Metric	Target
Time to generate proposal	< 2 minutes (including AI processing)
Human input required	< 1 minute (who, what, when)
Content accuracy	90%+ match with senior partner standards
Brand compliance	100% (Master Layout enforces brand)
Supported service types	FDD, TDD, IA, Valuation
Languages	Hungarian, English

Built with the Antigravity Intelligence Ecosystem DeckForge × Johanna × SlideMind × SlideForge